Macroeconomics PhD Intro
Lecture 4: Dynamic Competitive Equilibrium

Chapter 5: Arrow-Debreu, Sequential, Recursive, and OLG Equilibria

Swapnil Singh

Lietuvos Bankas

Kaunas University of Technology

Spring 2026

Learning Goals

- Define competitive equilibrium in dynamic economies: agents, prices, and market clearing.
- Compare three equilibrium languages: Arrow-Debreu, sequential, and recursive.
- State and derive core pricing and Euler conditions in endowment and production economies.
- Understand consumption smoothing and permanent-income results.
- Show why the neoclassical competitive equilibrium coincides with the planner allocation.
- See why overlapping-generations economies differ sharply from dynastic economies.

Road Map (Following Chapter 5)

1. Why dynamic competitive equilibrium matters.
2. Arrow-Debreu equilibrium: endowment, labor, and growth economies.
3. Sequential equilibrium: period-by-period trade.
4. Recursive equilibrium: steady state and dynamics.
5. Overlapping-generations economies and their distinctive implications.

Bridge from Lecture 3

Chapter 4 solved dynamic optimization problems for a single agent. Chapter 5 decentralizes them: many agents optimize simultaneously, and prices coordinate their choices through market clearing.

Why Move from Planner Problems to Market Equilibrium?

Chapter 4 solved intertemporal allocation for a planner or individual household. Macroeconomics, however, studies **decentralized economies**:

- households choose consumption, saving, and labor supply;
- firms choose factor demands and production;
- prices coordinate these choices across time and agents.

Central Question

Can decentralized optimization plus market clearing reproduce the planner allocation? Under perfect competition, often yes. Once frictions are introduced, often no. This distinction motivates everything from welfare theorems to policy analysis.

What Is a Dynamic Competitive Equilibrium?

A competitive equilibrium requires two sets of conditions:

1. **Individual optimality**: each household and firm solves its own problem, taking prices as given.
2. **Market clearing**: for every traded object, aggregate demand equals aggregate supply.

Objects Traded

Depending on the formulation:

- consumption goods at different dates,
- labor services (in efficiency units),
- capital services,
- one-period financial assets.

Key: prices are taken as given in each agent's problem. Variables not listed as choices are not optimized over—this is what "price-taking" means mathematically.

Perfect Competition as a Benchmark

- Perfect competition is a useful *approximation*, not an exact description.
- It captures how private incentives steer production and consumption through a **price mechanism**.
- Scarcity raises prices, which reduces demand and/or raises supply.
- Investment is influenced by the interest rate—the relative price of goods today vs. tomorrow.

Example: The Interest Rate

Consumers are more willing to forgo consumption and lend when the interest rate is high. Firms invest until the marginal return on investment equals the cost of borrowing. The interest rate equilibrates these two forces.

Three Ways to Define the Same Economy

Arrow-Debreu

- All trades at date 0.
- Goods dated by time.
- Equilibrium: price and quantity *sequences*.

Sequential

- Trading each period.
- Intertemporal trade through assets.
- Equilibrium: paths of quantities and spot/asset prices.

Recursive

- Sequential structure rewritten with dynamic programming.
- Equilibrium: *functions* of state variables.

Under standard conditions (complete markets, no frictions), all three formulations generate the **same allocations and prices**.

Which Equilibrium Concept Is Useful When?

- **Arrow-Debreu**: closest to general-equilibrium micro theory; clean for welfare and existence arguments.
- **Sequential**: closest to actual chronology; ideal when borrowing, lending, and period budgets matter. Generalizes easily to incomplete markets.
- **Recursive**: ideal for computation, policy functions, and state-based reasoning.

Important Distinction

Arrow-Debreu and sequential equilibria are typically defined *for a given initial state*. Recursive equilibrium is defined *for all relevant states*—the policy function applies to any (k, K).

Research Practice

Arrow-Debreu is conceptually useful, sequential is operationally convenient, and recursive is computationally indispensable.

Arrow-Debreu Idea: Date-0 Markets

In an Arrow-Debreu equilibrium, all trades are contracted at time 0.

- Every agent signs a **complete contract** at time 0 specifying deliveries at every future date.
- One unit of the consumption good at date t is a **distinct commodity**.
- Its price is p_t, measured in date-0 goods.
- We normalize $p_0 = 1$ (date-0 good is the **numéraire**).

Interpretation

The price sequence $\{p_t\}$ is a **term structure of intertemporal prices**: p_t tells you how much date-0 consumption you must sacrifice to obtain one unit of consumption at date t.

Enforcement: all promises are fully enforceable. Without this, markets are incomplete—a topic for later chapters.

Endowment Economy: Primitives

A continuum of consumers $i \in [0, 1]$ with **common** preferences

$$\sum_{t=0}^{\infty} \beta^t u(c_{i,t}), \tag{5.1}$$

and individual endowment sequence $\{y_{i,t}\}_{t=0}^{\infty}$.

Given Arrow-Debreu prices $\{p_t\}$, consumer i faces the **lifetime budget**

$$\sum_{t=0}^{\infty} p_t c_{i,t} = \sum_{t=0}^{\infty} p_t y_{i,t}. \tag{5.2}$$

- All intertemporal trade is collapsed into **one date-0 constraint**.
- Since utility is strictly increasing, the budget holds with equality.
- p_t/p_0 is the price of a time-t good in terms of time-0 consumption.

Definition: Arrow-Debreu Equilibrium (Endowment Economy)

Definition 1

An **Arrow-Debreu competitive equilibrium** is a set of sequences $\{c_{i,t}^\star\}_{t=0}^\infty$ for each $i \in [0,1]$, and $\{p_t\}_{t=0}^\infty$ such that

1. for each i, $\{c_{i,t}^\star\}$ solves

$$\max_{\{c_{i,t}\}_{t=0}^\infty} \sum_{t=0}^\infty \beta^t u(c_{i,t}) \quad \text{subject to} \quad \sum_{t=0}^\infty p_t c_{i,t} = \sum_{t=0}^\infty p_t y_{i,t};$$

2. for each t, goods market clearing holds:

$$\int_0^1 c_{i,t}^\star \, di = \int_0^1 y_{i,t} \, di \equiv Y_t.$$

Note: the definition lists what agents optimize over. Prices are *not* choice variables—agents take them as given.

Consumer FOCs and Intertemporal Prices

The Lagrangian for consumer i is

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t u(c_{i,t}) - \lambda_i \left[\sum_{t=0}^{\infty} p_t c_{i,t} - \sum_{t=0}^{\infty} p_t y_{i,t} \right].$$

First-order condition with respect to $c_{i,t}$:

$$\beta^t u'(c_{i,t}) = \lambda_i p_t.$$

- λ_i is the shadow value of lifetime wealth—**constant across time** but may differ across agents.
- With u strictly concave, this uniquely determines $c_{i,t}$ given λ_i and p_t.

Appendix

Eliminating the Multiplier: The Pricing Equation

At $t = 0$: $u'(c_{i,0}) = \lambda_i p_0 = \lambda_i$ (since $p_0 = 1$). Hence $\lambda_i = u'(c_{i,0})$.

Substitute into the date-t FOC:

$$p_t = \beta^t \frac{u'(c_{i,t})}{u'(c_{i,0})}. \tag{5.3}$$

Three Key Insights

1. p_t equals the **marginal rate of substitution** between date t and date 0 goods.
2. Since the *same* p_t applies to all households, **relative marginal utilities across dates are equalized across all households**.
3. With strictly concave u, this means consumption *growth rates* are identical across agents.

Log Utility: Closed-Form Prices

If $u(c) = \log c$, then $u'(c) = 1/c$, and

$$p_t = \beta^t \frac{c_{i,0}}{c_{i,t}}.$$

Multiply both sides by $c_{i,t}$ and integrate across agents:

$$p_t \underbrace{\int_0^1 c_{i,t}\, di}_{=Y_t} = \beta^t \underbrace{\int_0^1 c_{i,0}\, di}_{=Y_0} \quad \Rightarrow \quad \boxed{p_t = \beta^t \frac{Y_0}{Y_t}.}$$

Interpretation

- Prices fall over time due to **discounting** (β^t).
- Prices **rise** in periods with low aggregate endowment: marginal utility is high when resources are scarce.

Appendix

Consumption Growth and Perfect Smoothing

From the Euler condition, **all** consumers experience the same consumption growth:

$$\frac{c_{i,t+1}}{c_{i,t}} = \frac{c_{i',t+1}}{c_{i',t}} = \frac{Y_{t+1}}{Y_t} = \beta \frac{p_t}{p_{t+1}}.$$

- Aggregate endowment dynamics fully determine individual consumption growth rates.
- Idiosyncratic endowment risk mainly affects **levels**, not growth rates.

Special case: if aggregate endowment is constant ($Y_t = \bar{Y}$), then each consumer fully smooths consumption even when individual endowments fluctuate. This is the power of complete markets.

Consumption Levels and Permanent Income

Under log utility, combining the Euler equations and budget:

$$c_{i,0} \sum_{t=0}^{\infty} \beta^t = Y_0 \sum_{t=0}^{\infty} \beta^t \frac{y_{i,t}}{Y_t} \quad \Rightarrow \quad \boxed{c_{i,0} = (1-\beta)\, Y_0 \sum_{t=0}^{\infty} \beta^t \frac{y_{i,t}}{Y_t}.}$$

Interpretation

- Date-0 consumption is a fraction $(1-\beta)$ of "lifetime wealth."
- Endowments further into the future get lower weight (discounting).
- Endowments in **high-aggregate-income** periods get lower weight: resources are less valuable when everyone is rich.
- Two consumers with the same average endowment can have different wealth if one's endowments are high when others' are low.

Appendix

Consumption Shares Are Constant

Since consumption growth is identical for all consumers:

$$c_{i,t} = \theta_i \, Y_t, \qquad \theta_i = (1-\beta) \sum_{t=0}^{\infty} \beta^t \frac{y_{i,t}}{Y_t}.$$

- Each consumer's share of aggregate consumption is **constant over time**.
- If agents have identical endowments ($y_{i,t} = Y_t$), then $\theta_i = 1$ and we have a representative consumer.

CRRA Generalization

With $u(c) = \frac{c^{1-\sigma}-1}{1-\sigma}$, consumption growth rates are still identical:

$$\frac{c_{i,t+1}}{c_{i,t}} = \left(\beta \frac{p_t}{p_{t+1}}\right)^{1/\sigma}, \qquad p_t = \beta^t \left(\frac{Y_0}{Y_t}\right)^{\sigma}.$$

Higher curvature σ makes prices more sensitive to aggregate scarcity.

Representative-Agent Equilibrium (Special Case)

When all consumers have the same endowment, the equilibrium simplifies:

Definition 2

An **Arrow-Debreu competitive equilibrium** (representative agent) is $\{c_t^\star\}_{t=0}^\infty$ and $\{p_t\}_{t=0}^\infty$ such that

1. $\{c_t^\star\}$ solves $\max_{\{c_t\}} \sum_{t=0}^\infty \beta^t u(c_t)$ subject to $\sum_{t=0}^\infty p_t c_t = \sum_{t=0}^\infty p_t y_t$;
2. $c_t^\star = y_t$ for all t.

This definition is mathematically precise but the economic context—many identical consumers—is written "between the lines."

Production Economy with Labor: Setup

Households supply labor $\ell_{i,t}$ with disutility. Preferences:

$$\sum_{t=0}^{\infty} \beta^t \left[u(c_{i,t}) - v(\ell_{i,t}) \right]. \tag{5.4}$$

- $v(\ell)$: disutility of work—increasing and convex.
- e_i: efficiency units of labor (skills). Effective labor supplied $= e_i \ell_{i,t}$.

Technology: a representative firm with linear production

$$Y_t = A_t L_t, \tag{5.5}$$

where L_t is total effective labor input.

Firm's Problem Under CRS

The representative firm maximizes profits:

$$\max_{\{L_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} p_t(A_t L_t - w_t L_t) = \sum_{t=0}^{\infty} p_t(A_t - w_t) L_t. \tag{5.6}$$

Since L_t only appears in the time-t term, this reduces to a sequence of static problems.

Three Cases

- $A_t > w_t$: profits rise without bound with $L_t \Rightarrow$ no finite maximizer.
- $A_t < w_t$: negative profits $\Rightarrow$ unique maximizer is $L_t = 0$.
- $A_t = w_t$: $\pi_t = 0$ for every $L_t \geq 0 \Rightarrow$ **firm is indifferent over scale**.

$\Rightarrow$ Equilibrium requires $w_t = A_t$. Market clearing then pins down L_t.

Appendix

Definition: Arrow-Debreu Equilibrium (Labor Economy)

Definition 3

An **Arrow-Debreu competitive equilibrium** is $\{c^\star_{i,t}, \ell^\star_{i,t}\}$ for each i, $\{L^\star_t\}$, $\{p_t\}$, and $\{w_t\}$ such that

1. for each i, $\{c^\star_{i,t}, \ell^\star_{i,t}\}$ solves

$$\max_{\{c_t,\ell_t\}} \sum_{t=0}^{\infty} \beta^t [u(c_t) - v(\ell_t)] \text{ s.t. } \sum_{t=0}^{\infty} p_t c_t = \sum_{t=0}^{\infty} p_t w_t e_i \ell_t; \tag{5.7}$$

2. $\{L^\star_t\}$ solves $\max_{\{L_t\}} \sum_{t=0}^{\infty} p_t (A_t L_t - w_t L_t)$;
3. for each t: $\int_0^1 e_i \ell^\star_{i,t}\, di = L^\star_t$ and $\int_0^1 c^\star_{i,t}\, di = A_t L^\star_t$.

Market clearing operates on both labor (supply = demand) and goods (consumption = output).

Labor Economy: Household FOCs

The consumer's optimality conditions:

$$\beta^t u'(c_{i,t}) = \lambda_i p_t, \tag{5.8}$$

$$v'(\ell_{i,t}) = e_i w_t u'(c_{i,t}). \tag{5.9}$$

Intertemporal Condition (5.10)

Eliminate λ_i: $\dfrac{p_{t+1}}{p_t} = \dfrac{\beta u'(c_{i,t+1})}{u'(c_{i,t})}$. Same as the endowment economy.

Intratemporal Condition

Eq. (5.9) sets the **marginal disutility of work** equal to the **marginal utility value of wage income**. This determines labor supply for given consumption.

Appendix

Income and Substitution Effects on Labor Supply

With log utility ($u(c) = \log c$) and $w_t = A_t$:

$$v'(\ell_{i,t})\, c_{i,t} = e_i A_t.$$

Using $c_{i,t} = \theta_i Y_t$ and conjecturing $\theta_i = e_i$:

$$v'(\ell_{i,t})\, e_i\, A_t L_t = e_i A_t \quad \Rightarrow \quad v'(\ell_{i,t}) = \frac{1}{L_t}.$$

Result

All households supply the **same labor** $\ell_{i,t} = L_t$, regardless of skill e_i.

Why? An extra unit of work earns $e_i w_t$ (substitution effect: work more). But higher-skilled workers have higher consumption and lower marginal utility (income effect: work less). With log utility, the two **exactly cancel**.

Arrow-Debreu Neoclassical Growth Economy: Setup

Add capital to production:

$$Y_t = A_t K_t^{\alpha} L_t^{1-\alpha}. \tag{5.11}$$

Households own capital and rent it to firms. Capital accumulation:

$$k_{i,t+1} = (1-\delta)k_{i,t} + \iota_{i,t}. \tag{5.13}$$

Lifetime budget:

$$\sum_{t=0}^{\infty} p_t(c_{i,t} + \iota_{i,t}) = \sum_{t=0}^{\infty} p_t(r_t k_{i,t} + w_t e_i \ell_{i,t}). \tag{5.14}$$

Income sources: capital rental ($r_t k_{i,t}$) + labor income ($w_t e_i \ell_{i,t}$).
Expenditure: consumption ($c_{i,t}$) + investment ($\iota_{i,t}$).

Definition: Arrow-Debreu Equilibrium (Growth Economy)

Definition 4 (abbreviated)

An **Arrow-Debreu competitive equilibrium** is $\{c^\star_{i,t}, \iota^\star_{i,t}, \ell^\star_{i,t}\}$ for each i, $\{K^\star_t, L^\star_t\}$, and prices $\{p_t, p_t r_t, p_t w_t\}$ such that

1. for each i, $\{c^\star_{i,t}, \iota^\star_{i,t}, \ell^\star_{i,t}\}$ solves the household problem subject to budget (5.14) and capital accumulation (5.13);
2. $\{K^\star_t, L^\star_t\}$ solves $\max \sum_{t=0}^{\infty} p_t \left[A_t K_t^\alpha L_t^{1-\alpha} - r_t K_t - w_t L_t\right]$;
3. labor, capital, and goods markets clear:

$$\int e_i \ell^\star_{i,t}\, di = L^\star_t, \quad \int k^\star_{i,t}\, di = K^\star_t, \quad \int (c^\star_{i,t} + \iota^\star_{i,t})\, di = A_t (K^\star_t)^\alpha (L^\star_t)^{1-\alpha}.$$

Note: $k_{i,0}$ is exogenous and not an equilibrium object. $K_0 = \int k_{i,0}\, di$ is given.

Investment Pricing: A No-Arbitrage Condition

The consumer's optimality condition for investment is

$$\boxed{p_t = p_{t+1} r_{t+1} + (1-\delta) p_{t+1}.} \tag{5.17}$$

No-Arbitrage Interpretation

- **Cost** of investing one unit at date t: p_t (in date-0 goods).
- **Return** at $t+1$: rental income $p_{t+1} r_{t+1}$ plus undepreciated capital $(1-\delta) p_{t+1}$.
- Optimality $\Rightarrow$ cost $=$ return.

Dividing by p_{t+1}:

$$\frac{p_t}{p_{t+1}} = 1 - \delta + r_{t+1}.$$

The relative price of date-t consumption in terms of date-$(t+1)$ consumption equals the gross return on capital.

Appendix

Firm FOCs: Factor Prices Equal Marginal Products

The firm's problem is static date by date. Optimality gives:

$$r_t = \alpha A_t K_t^{\alpha-1} L_t^{1-\alpha} = F_1(K_t, L_t),$$

$$w_t = (1-\alpha) A_t K_t^{\alpha} L_t^{-\alpha} = F_2(K_t, L_t).$$

- Under CRS, for general (r_t, w_t) these two conditions may not hold simultaneously $\Rightarrow$ infinite profits or shutdown.
- In equilibrium, prices adjust so both hold, and the firm earns **zero profits**.
- The **capital–labor ratio** is uniquely pinned down; individual firm scale is not.

Euler's Theorem

With CRS: $F(K, L) = F_1 K + F_2 L = r_t K_t + w_t L_t$. Output is fully exhausted by factor payments.

Aggregate Equilibrium System (Log Utility)

Under log utility and the aggregation assumption $k_{i,0} = e_i K_0$, the equilibrium reduces to three aggregate equations:

$$\frac{C_{t+1}}{C_t} = \beta\Big(1 - \delta + \alpha A_{t+1} K_{t+1}^{\alpha-1} L_{t+1}^{1-\alpha}\Big), \tag{5.18}$$

$$v'(L_t) = \frac{(1-\alpha) A_t K_t^{\alpha} L_t^{-\alpha}}{C_t}, \tag{5.19}$$

$$C_t = A_t K_t^{\alpha} L_t^{1-\alpha} + (1-\delta) K_t - K_{t+1}. \tag{5.20}$$

- (5.18): Euler equation (intertemporal).
- (5.19): labor–leisure condition (intratemporal).
- (5.20): resource constraint.
- This is a second-order difference equation system in K—generally no closed-form solution.

Competitive Equilibrium = Planner Allocation

Assume identical labor efficiency, $e_i = 1$. The planner solves:

$$\max_{\{C_t, L_t, K_{t+1}\}} \sum_{t=0}^{\infty} \beta^t \left[\log C_t - v(L_t) \right]$$

subject to

$$C_t + K_{t+1} = A_t K_t^{\alpha} L_t^{1-\alpha} + (1-\delta) K_t.$$

Key Result

The planner's FOCs with respect to K_{t+1} and L_t are **exactly** equations (5.18) and (5.19). Therefore, the competitive equilibrium allocation equals the planner allocation.

This is the **First Welfare Theorem** in action. We will study it formally in Chapter 6. Appendix

Why Sequential Trade?

Arrow-Debreu is conceptually clean but unrealistic:

- No real economy has a single date-0 market for all future goods.
- Sequential equilibrium lets agents trade **period by period**.
- Intertemporal trade happens through **financial assets**.

In the endowment economy, the period budget constraint is

$$c_{i,t} + q_t a_{i,t+1} = a_{i,t} + y_{i,t}.$$

- $a_{i,t}$: asset holdings entering period t.
- q_t: price of one unit of the asset paying off at $t+1$.
- Gross interest rate: $1 + r_{t+1} = 1/q_t$.

New Infinite-Horizon Ingredient

We need a **no-Ponzi-game condition** to rule out agents borrowing indefinitely:

$$\lim_{t\to\infty} \left(\prod_{s=0}^{t} q_s \right) a_{i,t+1} \geq 0.$$

Definition: Sequential Equilibrium (Endowment Economy)

Definition 5

A **sequential competitive equilibrium** is $\{c^\star_{i,t}, a^\star_{i,t+1}\}$ for each i, and $\{q_t\}$ such that

1. for each i, $\{c^\star_{i,t}, a^\star_{i,t+1}\}$ solves

$$\max_{\{c_{i,t}, a_{i,t+1}\}} \sum_{t=0}^{\infty} \beta^t u(c_{i,t})$$

   subject to

$$c_{i,t} + q_t a_{i,t+1} = a_{i,t} + y_{i,t} \quad \forall t, \qquad a_0 = a_{i,0},$$

$$\lim_{t\to\infty} \left(\prod_{s=0}^{t} q_s\right) a_{i,t+1} \geq 0; \qquad \text{(nPg)}$$

2. for all t: $\displaystyle\int_0^1 c^\star_{i,t}\, di = Y_t$ and $\displaystyle\int_0^1 a^\star_{i,t+1}\, di = 0$.

Asset market clearing ($\int a_{i,t+1}\, di = 0$): agents borrow and lend to each other; trades net out.

Walras' Law

The definition imposes **two** market clearing conditions (goods and assets). But only one is independent.

Proof: integrate the household budget across i:

$$\underbrace{\int_0^1 c_{i,t}\, di}_{C_t} + q_t \underbrace{\int_0^1 a_{i,t+1}\, di}_{=0 \text{ if asset clears}} = \underbrace{\int_0^1 a_{i,t}\, di}_{=0} + \underbrace{\int_0^1 y_{i,t}\, di}_{Y_t}.$$

If asset market clears ($\int a_{i,t+1}\, di = 0$), then $C_t = Y_t$: goods market clears automatically.

Walras' Law

In an economy with n markets, if $n-1$ markets clear, the n-th must clear too. This is because budget constraints link all markets.

Sequential Euler Equation

With one budget constraint *per date*, there is one multiplier $\lambda_{i,t}$ per date:

$$\beta^t u'(c_{i,t}) = \lambda_{i,t}, \tag{5.21}$$

$$q_t \lambda_{i,t} = \lambda_{i,t+1}. \tag{5.22}$$

Eliminate multipliers:

$$\boxed{q_t = \beta \frac{u'(c_{i,t+1})}{u'(c_{i,t})}.}$$

Comparison with Arrow-Debreu

The economics is **unchanged**. The only difference is representation: sequential budgets generate time-varying multipliers $\lambda_{i,t}$ instead of one lifetime multiplier λ_i.

Appendix

Recovering Arrow-Debreu Prices from Sequential Trade

Multiply sequential Euler equations forward from 0 to $t-1$:

$$\prod_{s=0}^{t-1} q_s = \beta^t \frac{u'(c_{i,t})}{u'(c_{i,0})}.$$

The left-hand side is exactly the Arrow-Debreu date-t price:

$$\boxed{p_t = \prod_{s=0}^{t-1} q_s.}$$

Hence

$$\frac{p_{t-1}}{p_t} = \frac{1}{q_{t-1}} = 1 + r_t.$$

Interpretation

Arrow-Debreu prices are **present values**: the product of one-period asset prices. The gross interest rate between $t-1$ and t is the ratio of consecutive Arrow-Debreu prices.

Appendix

Sequential Budgets Collapse to the Lifetime Budget

Starting from $c_{i,t} + q_t a_{i,t+1} = a_{i,t} + y_{i,t}$, substitute forward repeatedly:

$$\sum_{t=0}^{T} \underbrace{\left(\prod_{s=0}^{t-1} q_s\right)}_{=p_t} c_{i,t} + \left(\prod_{s=0}^{T-1} q_s\right) q_T \, a_{i,T+1} = a_{i,0} + \sum_{t=0}^{T} p_t \, y_{i,t}.$$

Take $T \to \infty$ and use the nPg/TVC condition:

$$\sum_{t=0}^{\infty} p_t c_{i,t} \leq a_{i,0} + \sum_{t=0}^{\infty} p_t y_{i,t}.$$

With strictly increasing utility, the inequality binds (TVC holds with equality). Sequential and Arrow-Debreu budget sets **coincide**.

Appendix

Definition: Sequential Equilibrium (Growth Economy)

Definition 6

A **sequential competitive equilibrium** is $\{c^\star_{i,t}, k^\star_{i,t+1}\}$ for each i, and $\{r_t, w_t\}$ such that

1. for each i, $\{c^\star_{i,t}, k^\star_{i,t+1}\}$ solves

$$\max \sum_{t=0}^{\infty} \beta^t u(c_{i,t}) \text{ s.t. } c_{i,t} + k_{i,t+1} = (1 - \delta + r_t)k_{i,t} + w_t e_i,$$

   with $k_{i,0}$ given, plus the nPg condition;
2. for each t, $(K_t, 1)$ solves $\max_{K,L} A_t K^\alpha L^{1-\alpha} - r_t K - w_t L$;
3. for each t: $C_t + K_{t+1} = A_t K_t^\alpha + (1 - \delta)K_t$.

Walras' Law: the resource constraint (3) is redundant given (1) and (2)—summing household budgets and using the firm's CRS problem delivers it.

Why Sequential Equilibrium Is Often the Workhorse

- It makes borrowing, lending, and financial assets **explicit**.
- Period budgets map naturally into Euler equations and transversality conditions.
- It generalizes easily to **incomplete markets** and borrowing constraints.
- It retains full equivalence with Arrow-Debreu when markets are complete.

Summary

Feature	**Arrow-Debreu**	**Sequential**
Budget	one lifetime	one per period
Multiplier	λ_i (single)	$\lambda_{i,t}$ (one per t)
Prices	$\{p_t\}$	$\{q_t\}$ or $\{r_t, w_t\}$
Link	$p_t = \prod_{s=0}^{t-1} q_s$	$q_t = p_{t+1}/p_t$

Why Recursive Equilibrium?

Recursive equilibrium rewrites a sequential economy using **state variables** and **functions**:

- Sequences $\rightarrow$ policy rules $g(k, K)$.
- Time-dated prices $\rightarrow$ pricing functions $r(K), w(K)$.
- Agents take pricing functions and aggregate laws of motion as given.
- Equilibrium requires these functions to be **mutually consistent**.

State Variable Criterion

A variable is a state if it is:

1. **relevant** for current and future choices, and
2. **predetermined** at the start of the period.

Example: capital is a state; labor supply is not (it is chosen within the period).

Recursive Steady State: Endowment Economy

With constant endowment y_i, define the Bellman equation:

$$V_i(a) = \max_{a'} \; u(a + y_i - qa') + \beta V_i(a').$$

Definition 7

A **recursive steady-state competitive equilibrium** is a scalar q, functions $V_i^\star(a)$ and $g_i^\star(a)$, and asset holdings $a_i^\star$ for each i such that

1. for each i, $V_i^\star$ solves the Bellman equation, with $g_i^\star(a)$ attaining the max $\forall a$;
2. for each i, $g_i^\star(a_i^\star) = a_i^\star$ (stationarity);
3. $\int_0^1 g_i^\star(a_i^\star)\, di = 0$ (market clearing).

Note: the definition specifies *functions* (for all a), not just equilibrium values. Agents consider deviations from steady state but *choose* to stay.

Appendix

Characterizing the Recursive Steady State (Endowment)

The functional Euler equation:

$$q\, u'(a + y_i - qg_i(a)) = \beta u'\big(g_i(a) + y_i - qg_i(g_i(a))\big).$$

At the stationary point $a_i^\star = g_i^\star(a_i^\star)$:

$$q\, u'(a_i^\star(1-q) + y_i) = \beta u'(a_i^\star(1-q) + y_i).$$

Hence $q = \beta$ and $g_i^\star(a) = a$ for all a and i.

Permanent Income

Each agent preserves wealth and consumes permanent income: $c_i = a_i(1-\beta) + y_i$. The asset distribution is indeterminate—any distribution summing to zero is an equilibrium.

Recursive Steady State: Neoclassical Growth Economy

With exogenous labor e_i and constant TFP:

Definition 8

A **recursive steady-state competitive equilibrium** is scalars r, w, functions $V_i^\star(k), g_i^\star(k)$, and capital holdings $k_i^\star$ for each i such that

1. for each i, $V_i^\star$ solves $V_i(k) = \max_{k'} u((1-\delta+r)k + we_i - k') + \beta V_i(k')\ \forall k$;
2. $(K^\star, 1)$ solves $\max_{K,L} F(K,L) - rK - wL$, with $\int k_i^\star\, di = K^\star$;
3. for each i, $g_i^\star(k_i^\star) = k_i^\star$ (stationarity).

Steady-State Pricing

By stationarity: $1 = \beta(1-\delta+r)$. Then $r = F_1(K^\star, 1)$ pins down $K^\star$, and $w = F_2(K^\star, 1)$.
The long-run interest rate is determined by **impatience** and **depreciation** alone.

Identifying the State in the Dynamic Growth Economy

Consider the neoclassical model with a representative consumer.

At time t, current prices are determined by the aggregate capital stock:

$$r_t = F_1(K_t, 1), \qquad w_t = F_2(K_t, 1).$$

Therefore aggregate capital K is the relevant aggregate state.

For an individual household, own capital k matters too. Hence the recursive problem depends on (k, K).

Why Only K?

Think about what determines prices: the marginal products depend on the capital–labor ratio. With fixed labor, only K matters. The full *distribution* of capital across agents does not matter for prices when utility is CRRA—this is the aggregation result.

Definition: Recursive Competitive Equilibrium (Dynamics)

The household Bellman equation:

$$V(k, K) = \max_{k'} \; u\big((1 - \delta + r(K))k + w(K) - k'\big) + \beta V\big(k', G(K)\big). \tag{5.23}$$

Definition 9

A **recursive competitive equilibrium** is functions $r(K), w(K), G^\star(K), V^\star(k, K), g^\star(k, K)$ such that

1. $V^\star$ solves (5.23) with $G = G^\star$, and $g^\star(k, K)$ attains the max;
2. for all K: $r(K) = F_1(K, 1)$ and $w(K) = F_2(K, 1)$;
3. **consistency**: $G^\star(K) = g^\star(K, K)$ for all K.

Condition 3 is the key new element: the aggregate law of motion must be **consistent** with individual optimization when each agent holds $k = K$.

Appendix

Consistency: Why $G(K) = g(K, K)$?

- Households solve for $g(k, K)$: "if I have k and the economy has K, I save $g(k, K)$."
- The economy posits an aggregate law of motion $K' = G(K)$.
- In a representative-agent economy, each household enters with $k = K$.
- So aggregate capital next period, as chosen by households, is $g(K, K)$.
- This must equal the posited $G(K)$: a **fixed-point condition**.

Resource Constraint Is Automatic

At $k = K$: consumption $= (1 - \delta + r(K))K + w(K) - G(K)$. Using CRS:
$= F(K, 1) + (1 - \delta)K - G(K)$, which is output minus investment.

Functional Euler Equation

From the Bellman FOC and envelope theorem:

$$\begin{aligned} u'\big((1-\delta+r(K))k+w(K)-g(k,K)\big) = \beta\, u'\big(&g(k,K)(1-\delta+r(G(K))) \\ &+ w(G(K)) - g(g(k,K),G(K))\big) \\ &\times \big(1-\delta+r(G(K))\big). \end{aligned}$$

- This determines the policy function $g(k,K)$ given r, w, and G.
- **Evaluate at** $k=K$ and use CRS: the resulting equation *coincides* with the planner's recursive Euler equation from Chapter 4.
- Generally must be solved **numerically** (except with log utility, Cobb-Douglas F, and $\delta=1$).

Appendix

Aggregation Under Power Utility

When utility is CRRA, individual saving is **linear** in own wealth:

$$g_i(k, K) = \mu_i(K) + \lambda(K)\, k.$$

- The slope $\lambda(K)$ is **common across agents**—marginal propensity to save is the same.
- The intercept $\mu_i(K)$ varies with individual labor productivity e_i.
- Therefore aggregate saving depends only on K, not on the full wealth distribution.
- This is why **representative-agent aggregation** works in the benchmark dynastic model.

Important Limit

With **endogenous labor**, aggregation can fail unless the ratio of initial capital to labor productivity is the same across agents ($k_{i,0} = e_i K_0$). Otherwise, the wealth distribution becomes a relevant state variable.

Recursive Equilibrium with Valued Leisure

If households choose labor too, the Bellman problem becomes

$$V(k, K) = \max_{k', \ell} \; u\big((1 - \delta + r(K))k + w(K)\ell - k'\big) - v(\ell) + \beta V(k', G(K)).$$

Definition 10

Add to Definition 9:

- functions $H^\star(K)$ (aggregate labor supply) and $h^\star(k, K)$ (individual labor rule);
- pricing: $r(K) = F_1(K, H^\star(K))$, $w(K) = F_2(K, H^\star(K))$;
- consistency: $G^\star(K) = g^\star(K, K)$ and $H^\star(K) = h^\star(K, K)$.

Why Labor Is Not a State

Current labor supply is chosen within the period. It affects prices, but it is **not predetermined**. The state remains (k, K).

Why OLG Economies Are Different

In an overlapping-generations economy:

- the economy lasts **forever**,
- each individual lives only a **finite number of periods**,
- in the benchmark, parents are **not altruistic** toward children,
- **age** (not dynasty) is the core source of heterogeneity.

Key Implications

1. The competitive equilibrium may **fail to be Pareto efficient**.
2. Long-run interest rates are **not simply** determined by β and δ.
3. Dynamics follow a **first-order** (not second-order) difference equation.
4. Government transfers (e.g., PAYG pensions) can **improve welfare**.

Two-Period Life OLG: Setup

Each cohort born at t lives two periods (young at t, old at $t+1$):

$$u_t(c_y, c_o) = u(c_y) + \beta u(c_o).$$

Cohort $t = -1$ (initial old): $u_{-1}(c) = u(c)$.

Endowments: $(\omega_{y,t}, \omega_{o,t+1})$ for cohort t.

Budget constraints for cohort t:

$$c_{y,t} + q_t a_{t+1} = \omega_{y,t}, \qquad c_{o,t+1} = \omega_{o,t+1} + a_{t+1}.$$

- Young save (or borrow) through a one-period asset.
- Old consume endowment plus asset payoff.
- No bequest motive in the benchmark.

Definition: Sequential Equilibrium (OLG Endowment)

Definition 11

A **sequential competitive equilibrium** is $\{c^\star_{y,t}, c^\star_{o,t+1}, a^\star_{t+1}\}_{t\geq 0}$ and $\{q_t\}_{t\geq 0}$ such that

1. for each $t \geq 0$, $(c^\star_{y,t}, c^\star_{o,t+1}, a^\star_{t+1})$ solves

$$\max_{c_y, c_o, a'} u(c_y) + \beta u(c_o) \text{ s.t. } c_y + q_t a' = \omega_{y,t},\ c_o = \omega_{o,t+1} + a';$$

   and $c^\star_{o,0} = \omega_{o,0}$;
2. for all $t \geq 0$: $c^\star_{y,t} + c^\star_{o,t} = \omega_{y,t} + \omega_{o,t}$.

Key difference from dynastic: each generation's problem is a *finite*-horizon problem—no transversality or no-Ponzi condition needed.

OLG Endowment Equilibrium Is Autarky

Goods market clearing at date t:

$$\underbrace{c_{y,t}}_{\text{young at } t} + \underbrace{c_{o,t}}_{\text{old at } t} = \omega_{y,t} + \omega_{o,t}.$$

Using the young's budget ($c_{y,t} = \omega_{y,t} - q_t a_{t+1}$) and the old's budget ($c_{o,t} = \omega_{o,t} + a_t$), and noting that only the cohort-t young and cohort-$(t-1)$ old are alive:

$\Rightarrow$ market clearing implies $a^\star_{t+1} = 0$ for all t.

Autarky Result

$$c^\star_{y,t} = \omega_{y,t}, \qquad c^\star_{o,t} = \omega_{o,t}.$$

Each generation consumes its own endowment. There is **no intergenerational trade** because the young cannot trade with unborn future generations, and the old cannot commit future resources.

Appendix

OLG Prices and Negative Real Rates

Evaluate the Euler equation at autarky:

$$q_t u'(\omega_{y,t}) = \beta u'(\omega_{o,t+1}).$$

With CRRA utility $u(c) = \frac{c^{1-\sigma}-1}{1-\sigma}$:

$$\boxed{q_t = \beta \left(\frac{\omega_{y,t}}{\omega_{o,t+1}} \right)^{\sigma}.} \tag{5.24}$$

- If $\omega_y > \omega_o$ (income is front-loaded), then q_t can exceed 1.
- Since $1 + r_{t+1} = 1/q_t$, this means the **real interest rate can be negative**.
- This is *impossible* in the dynastic model with constant aggregate endowment.

Intuition

When life-cycle income declines with age and there is no intergenerational trade, future consumption is relatively scarce and therefore highly valued.

Why OLG Equilibrium May Be Inefficient

Suppose the young are relatively rich and the old relatively poor: $\omega_y > \omega_o$.

A **transfer from young to old** can raise welfare of the current old. If the transfer is repeated each period (a "social contract"):

- Current old: receive transfer, strictly better off.
- Current young: give transfer now, but receive one when old.
- Net effect on the young depends on the interest rate vs. the "biological return" (population growth rate).

Pay-As-You-Go Pensions

A government-run PAYG pension system implements exactly this transfer. It can **Pareto-improve** upon the market allocation in the OLG economy.

This is fundamentally different from the dynastic model, where the competitive equilibrium is always Pareto efficient (First Welfare Theorem).

Definition: Sequential Equilibrium (OLG Growth Economy)

Definition 12

A **sequential competitive equilibrium** is $\{c^\star_{y,t}, c^\star_{o,t+1}, k^\star_{t+1}\}$, $\{r_t, w_t\}$ s.t.

1. for each t, $(c^\star_{y,t}, c^\star_{o,t+1}, k^\star_{t+1})$ solves

$$\max u(c_y) + \beta u(c_o) \text{ s.t. } c_y + k' = w_t e_y,\ c_o = w_{t+1} e_o + (1 - \delta + r_{t+1}) k';$$

   and $c^\star_{o,0} = w_0 e_o + (1 - \delta + r_0) k_0$;
2. $r_t = F_1(k^\star_t, 1)$, $w_t = F_2(k^\star_t, 1)$, with $k^\star_0 \equiv k_0$;
3. $c^\star_{y,t} + c^\star_{o,t} + k^\star_{t+1} = F(k^\star_t, 1) + (1 - \delta) k^\star_t$.

Young save by **accumulating capital**: $k_{t+1} = w_t e_y - c_{y,t}$.
Old consume labor income plus capital returns: $c_{o,t+1} = w_{t+1} e_o + (1 - \delta + r_{t+1}) k_{t+1}$.

OLG Growth: Euler Equation and Dynamics

Substituting prices as functions of capital:

$$u'\big(e_y F_2(k_t, 1) - k_{t+1}\big) = \beta\, u'\big(e_o F_2(k_{t+1}, 1) + \big(1 - \delta + F_1(k_{t+1}, 1)\big) k_{t+1}\big) \times \big(1 - \delta + F_1(k_{t+1}, 1)\big).$$

First-Order Dynamics

Given k_t, this equation directly solves for k_{t+1}. The OLG model gives a **first-order** difference equation, unlike the dynastic model's **second-order** system.

"Solve forward" property

Start with k_0, solve for k_1, then k_2, and so on. No saddle-path selection is needed.

Appendix

OLG Steady State Is Not Dynastic

The steady state $\bar{k}$ satisfies:

$$u'\big(e_y F_2(\bar{k},1) - \bar{k}\big) = \beta\, u'\big(e_o F_2(\bar{k},1) + (1-\delta+F_1(\bar{k},1))\bar{k}\big)\big(1-\delta+F_1(\bar{k},1)\big).$$

Contrast with Dynastic Steady State

Dynastic: $1 = \beta(1-\delta+F_1(\bar{k},1)) \Rightarrow$ interest rate pinned by β and δ alone.
OLG: depends on utility curvature, life-cycle income (e_y, e_o), and the production function. The gross interest rate can be above or below $1/\beta$.

Exercise

With log u, $e_y = 1$, $e_o = 0$, Cobb-Douglas F, and $\delta = 1$: the steady-state gross interest rate is $\alpha(1+\beta)/(1-\alpha)$, which is less than 1 if α is low enough.

Warm Glow, Altruism, and the Road Back to Dynasties

Two important extensions:

1. **Perpetual youth (Blanchard-Yaari):**
   - All individuals face a constant probability of death.
   - As survival probability $\to 1$, the economy approaches dynastic behavior.
2. **Warm glow bequests:**

$$u(c_y) + \beta u(c_o) + \varphi(b').$$

Old agents get utility from the **act of giving** itself, not from the child's utility.

3. **Pure altruism:** parents value $\varphi(b')$ where φ is the child's *indirect utility*:

$$\varphi(b) = \max_{c_y, c_o, a', b'} u(c_y) + \beta u(c_o) + \tilde{\beta}\,\varphi(b').$$

This generates a **recursive dynasty**: the long-run interest rate is $1/\tilde{\beta}$.

Takeaway

As bequests become sufficiently dynastic, the long-run interest-rate logic converges to the benchmark immortal-family model.

Comparing the Three Equilibrium Languages

Arrow-Debreu

- All trade at date 0.
- Solve for price and allocation sequences.
- Best for theory and welfare.

Sequential

- Trade every period.
- Assets implement intertemporal trade.
- Best for chronology and finance.

Recursive

- Everything as functions of states.
- Solve Bellman plus consistency.
- Best for computation and dynamics.

	A-D	**Sequential**	**Recursive**
Equilibrium object	sequences	sequences	functions
Budget	lifetime	per-period	Bellman
Welfare analysis	✓✓	✓	✓
Computation	–	✓	✓✓
Incomplete mkts	–	✓✓	✓✓

OLG vs. Dynastic: Summary of Differences

Feature	**Dynastic**	**OLG**
Horizon	infinite	finite per agent
Dynamics	2nd-order	1st-order
SS interest rate	$\beta(1-\delta+r)=1$	depends on many parameters
Negative real rates	generally no	possible
Pareto efficient	yes (under FWT)	not necessarily
Role for gov't transfers	none (in benchmark)	PAYG pensions can help
Saddle-path selection	yes	no ("solve forward")

What Chapter 5 Delivers

- Competitive equilibrium = optimization + market clearing, defined carefully for dynamic economies.
- Arrow-Debreu, sequential, and recursive formulations are **different languages** for the same benchmark allocation.
- In the neoclassical dynastic model, equilibrium **decentralizes the planner solution**.
- In OLG economies, that equivalence **breaks down**: interest rates, dynamics, and welfare properties all differ.
- Aggregation works under CRRA preferences (with caveats for endogenous labor).

Big Picture

Chapter 5 provides the market benchmark. Chapter 6 asks when that benchmark is socially efficient—and when it is not.

Next Lecture (Chapter 6)

We turn from equilibrium *characterization* to equilibrium *welfare*:

- Pareto optimality and the First Welfare Theorem in an abstract setting.
- Mapping the FWT to our specific macroeconomic environments.
- Why competitive equilibria can fail to be efficient: externalities, market power, incomplete markets, public goods.
- The role of government policy in correcting market failures.

A1. Arrow-Debreu Price Formula from the Consumer FOC

The Lagrangian for consumer i:

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t u(c_{i,t}) - \lambda_i \Big[\sum_{t=0}^{\infty} p_t c_{i,t} - \sum_{t=0}^{\infty} p_t y_{i,t} \Big].$$

FOC w.r.t. $c_{i,t}$:

$$\beta^t u'(c_{i,t}) = \lambda_i p_t.$$

At $t = 0$: $u'(c_{i,0}) = \lambda_i p_0 = \lambda_i$ (since $p_0 = 1$).
Substitute $\lambda_i = u'(c_{i,0})$ into the date-t FOC:

$$p_t = \beta^t \frac{u'(c_{i,t})}{u'(c_{i,0})}.$$

Key: this holds for every i, so $\beta^t u'(c_{i,t})/u'(c_{i,0})$ is the same across all consumers.

Back

A2. Log Utility and Aggregate Endowment Pricing

With $u(c) = \log c$, the pricing condition becomes

$$p_t = \beta^t \frac{c_{i,0}}{c_{i,t}}.$$

Multiply by $c_{i,t}$ and integrate across agents:

$$p_t \int_0^1 c_{i,t}\, di = \beta^t \int_0^1 c_{i,0}\, di.$$

Use market clearing:

$$\int_0^1 c_{i,t}\, di = Y_t, \qquad \int_0^1 c_{i,0}\, di = Y_0.$$

Hence

$$p_t = \beta^t \frac{Y_0}{Y_t}.$$

Note: This "trick" of multiplying by $c_{i,t}$ and integrating works because log utility makes $p_t c_{i,t} = \beta^t c_{i,0}$ *linear* in date-0 consumption.

Back

A3. Investment Condition as No-Arbitrage

One additional unit of investment at date t costs one unit of the good, valued at p_t in date-0 units.

At $t+1$ it delivers:

- rental payment: $p_{t+1} r_{t+1}$;
- undepreciated capital: $(1-\delta)p_{t+1}$.

Optimality requires cost = return:

$$p_t = p_{t+1} r_{t+1} + (1-\delta)p_{t+1}.$$

Alternative derivation: from the household Lagrangian, substitute $\iota_{i,t} = k_{i,t+1} - (1-\delta)k_{i,t}$ into the budget. Take FOC w.r.t. $k_{i,t+1}$:

$$-p_t + p_{t+1} r_{t+1} + (1-\delta)p_{t+1} = 0.$$

Back

A4. Planner FOCs Match Competitive Equilibrium

Planner problem:

$$\max \sum_{t=0}^{\infty} \beta^t \left[\log C_t - v(L_t) \right] \quad \text{s.t.} \quad C_t + K_{t+1} = A_t K_t^{\alpha} L_t^{1-\alpha} + (1-\delta) K_t.$$

Substitute the constraint and take FOCs.

FOC w.r.t. K_{t+1}:

$$\frac{1}{C_t} = \beta \frac{1}{C_{t+1}} \left(1 - \delta + \alpha A_{t+1} K_{t+1}^{\alpha-1} L_{t+1}^{1-\alpha} \right).$$

FOC w.r.t. L_t:

$$v'(L_t) = \frac{(1-\alpha) A_t K_t^{\alpha} L_t^{-\alpha}}{C_t}.$$

These are **identical** to the competitive equilibrium conditions (5.18) and (5.19).

Back

A5. Sequential Euler Equation from Time-Dated Multipliers

Sequential household problem:

$$\max \sum_{t=0}^{\infty} \beta^t u(c_t) \quad \text{s.t.} \quad c_t + q_t a_{t+1} = a_t + y_t \;\; \forall t.$$

Lagrangian: $\mathcal{L} = \sum_{t=0}^{\infty} \left[\beta^t u(c_t) - \lambda_t (c_t + q_t a_{t+1} - a_t - y_t)\right]$.

FOCs:

$$\frac{\partial \mathcal{L}}{\partial c_t}: \quad \beta^t u'(c_t) = \lambda_t,$$

$$\frac{\partial \mathcal{L}}{\partial a_{t+1}}: \quad q_t \lambda_t = \lambda_{t+1}.$$

Substitute the first into the second:

$$q_t \beta^t u'(c_t) = \beta^{t+1} u'(c_{t+1}) \quad \Rightarrow \quad q_t = \beta \frac{u'(c_{t+1})}{u'(c_t)}.$$

Back

A6. Rolling Sequential Budgets into the Lifetime Budget

At $t = 0$ and $t = 1$:

$$c_0 + q_0 a_1 = a_0 + y_0, \qquad c_1 + q_1 a_2 = a_1 + y_1.$$

From the first equation: $a_1 = (a_0 + y_0 - c_0)/q_0$. Substitute into the second:

$$c_0 + q_0 c_1 + q_0 q_1 a_2 = a_0 + y_0 + q_0 y_1.$$

Continue forward and use $p_t = \prod_{s=0}^{t-1} q_s$:

$$\sum_{t=0}^{T} p_t c_t + p_T q_T a_{T+1} = a_0 + \sum_{t=0}^{T} p_t y_t.$$

Take $T \to \infty$; the nPg/TVC condition forces $\lim p_T q_T a_{T+1} \geq 0$; strict monotonicity of u makes it bind at 0.

Back

A7. Recursive Steady State in the Endowment Economy

Bellman equation:

$$V_i(a) = \max_{a'} u(a + y_i - qa') + \beta V_i(a').$$

Functional Euler equation (FOC + envelope):

$$q\, u'(a + y_i - qg_i(a)) = \beta u'\big(g_i(a) + y_i - qg_i(g_i(a))\big).$$

Evaluate at a stationary point $a_i^\star = g_i(a_i^\star)$:

$$q\, u'(a_i^\star(1-q) + y_i) = \beta u'(a_i^\star(1-q) + y_i).$$

Thus $q = \beta$. Then the Euler equation implies $a - qg_i(a) = g_i(a) - qg_i(g_i(a))$, and $g_i^\star(a) = a$ solves this for all i.

Value function: $V_i^\star(a) = u(a(1-\beta) + y_i)/(1-\beta)$.

Back

A8. Consistency in Recursive Equilibrium

Households solve for an individual policy rule $g(k, K)$, taking as given:

$$r(K), \quad w(K), \quad G(K).$$

In a representative-agent equilibrium, each household enters with $k = K$, so next-period aggregate capital chosen by households is $g(K, K)$.

Equilibrium also posits an aggregate law of motion $K' = G(K)$. Therefore consistency requires

$$G(K) = g(K, K) \quad \forall K.$$

Interpretation: If you told each household "aggregate capital will evolve as $G(K)$" and they optimized accordingly, their actual collective behavior must *confirm* the law of motion G.

Back

A9. Functional Euler Equation in the Recursive Growth Model

Bellman equation:

$$V(k, K) = \max_{k'} u(c) + \beta V(k', G(K)),$$

where $c = (1 - \delta + r(K))k + w(K) - k'$.

FOC:

$$u'(c) = \beta V_1(k', G(K)).$$

Envelope theorem (differentiate value function w.r.t. k):

$$V_1(k, K) = u'(c)(1 - \delta + r(K)).$$

Advance one period:

$$V_1(k', G(K)) = u'(c')\big(1 - \delta + r(G(K))\big).$$

Substituting gives the functional Euler equation in the main text.

Back

A10. Why the OLG Endowment Equilibrium Is Autarky

At any date t, only two cohorts are alive:

- **Young** (cohort t): budget $c_{y,t} + q_t a_{t+1} = \omega_{y,t}$.
- **Old** (cohort $t-1$): budget $c_{o,t} = \omega_{o,t} + a_t$.

Goods market clearing:

$$c_{y,t} + c_{o,t} = \omega_{y,t} + \omega_{o,t}.$$

Substitute budgets:

$$(\omega_{y,t} - q_t a_{t+1}) + (\omega_{o,t} + a_t) = \omega_{y,t} + \omega_{o,t}.$$

$\Rightarrow a_t = q_t a_{t+1}$. Starting from $a_0 = 0$ (initial old have no assets from trade), iterate forward: $a_{t+1} = 0$ for all t.

$\Rightarrow c^\star_{y,t} = \omega_{y,t}$ and $c^\star_{o,t} = \omega_{o,t}$: **autarky**.

Back

A11. OLG Steady State Is Not Dynastic

Steady state $\bar{k}$ in the OLG growth model satisfies:

$$u'\big(e_y F_2(\bar{k},1) - \bar{k}\big) = \beta\, u'\big(e_o F_2(\bar{k},1) + (1-\delta+F_1(\bar{k},1))\bar{k}\big)\big(1-\delta+F_1(\bar{k},1)\big).$$

Contrast with dynastic steady state:

$$1 = \beta(1-\delta+F_1(\bar{k},1)).$$

The dynastic condition pins $F_1(\bar{k},1)$ independently of utility curvature or life-cycle income. The OLG condition involves the full structure of preferences and endowments.

Back

A12. Why the Firm's Optimal Scale Is Not Unique Under CRS

In the labor-only production economy, the firm's date-t problem is

$$\max_{L_t \geq 0} \pi_t(L_t) = A_t L_t - w_t L_t = (A_t - w_t) L_t.$$

Three cases:

- $A_t > w_t$: π_t rises linearly with $L_t \Rightarrow$ no finite maximizer.
- $A_t < w_t$: negative profits for any $L_t > 0 \Rightarrow$ unique maximizer is $L_t = 0$.
- $A_t = w_t$: $\pi_t = 0$ for every $L_t \geq 0 \Rightarrow$ firm is indifferent over scale.

Equilibrium Implication

CRS + price-taking forces $w_t = A_t$. Profit maximization pins down *zero profits*, not a unique scale. Market clearing then selects L_t.

Back

A13. Consumption Levels under Log Utility

From the Euler equations: $p_t c_{i,t} = \beta^t c_{i,0}$ for all t (since $p_t = \beta^t c_{i,0}/c_{i,t}$).

Substitute into the budget constraint:

$$\sum_{t=0}^{\infty} p_t c_{i,t} = \sum_{t=0}^{\infty} \beta^t c_{i,0} = \frac{c_{i,0}}{1-\beta} = \sum_{t=0}^{\infty} p_t y_{i,t}.$$

Solving:

$$c_{i,0} = (1-\beta) \sum_{t=0}^{\infty} p_t y_{i,t} = (1-\beta) Y_0 \sum_{t=0}^{\infty} \beta^t \frac{y_{i,t}}{Y_t}.$$

The last step uses $p_t = \beta^t Y_0 / Y_t$. This is the **permanent income** formula: consume a fraction $(1-\beta)$ of your present-value wealth, where wealth is measured in the "natural" units given by the equilibrium price sequence.

Back

A14. Intratemporal Condition: Labor–Leisure

From the household Lagrangian in the labor economy, the FOC w.r.t. $\ell_{i,t}$:

$$\frac{\partial}{\partial \ell_{i,t}}: \quad -\beta^t v'(\ell_{i,t}) + \lambda_i p_t w_t e_i = 0.$$

Using $\beta^t u'(c_{i,t}) = \lambda_i p_t$ (the consumption FOC):

$$v'(\ell_{i,t}) = e_i w_t u'(c_{i,t}).$$

Reading the Condition

- LHS: marginal cost of working (disutility).
- RHS: marginal benefit = wage $\times$ skill $\times$ marginal utility of the consumption purchased.
- This is a **static** condition within each period—no intertemporal dimension.

Back

A15. Detailed Budget Consolidation (Sequential $\to$ Lifetime)

Start with period budgets and define $p_0 \equiv 1$, $p_t \equiv \prod_{s=0}^{t-1} q_s$:
Step 1. $t = 0$: $c_0 + q_0 a_1 = a_0 + y_0$. Multiply by $p_0 = 1$:

$$p_0 c_0 + p_1 a_1 = a_0 + p_0 y_0.$$

Step 2. $t = 1$: $c_1 + q_1 a_2 = a_1 + y_1$. Multiply by $p_1 = q_0$:

$$p_1 c_1 + p_2 a_2 = p_1 a_1 + p_1 y_1.$$

Step 3. Add Steps 1 and 2; $p_1 a_1$ cancels:

$$p_0 c_0 + p_1 c_1 + p_2 a_2 = a_0 + p_0 y_0 + p_1 y_1.$$

General: after T iterations, the "residual" term is $p_{T+1} a_{T+1}$. As $T \to \infty$, the nPg condition forces this ≥ 0; optimality (TVC) forces it $= 0$.

Back